ASSIGNMENT ONE (10Marks)

To be submitted before 15/05/2025 10.00AM

Question One

Study the queuing model II and solve the following problems

a) A Commercial bank has 3 cash paying assistants customers are found to arrive in a Poisson fashion at an average rate of 6/hr for business transaction. The service time is found to have an exponential a distribution with a mean of 18 mins. Identify the queuing model and Calculate:
    i. Average number of customers in the system
    ii. Average time a customer spends in the system
    iii. Average queue length
    iv. How many hours a week can a cash paying assistant spend with the customers.

b) A one person barbershop has six chairs to accommodate people waiting for hair cut. Assume customers who arrive when all six chairs are full, leave without entering the barbershop. Customers arrive at the average rate of 3 per hour and spend on average of 15minutes in the barbershop. Identify the queuing model then find the
    i. Probability a customer can get directly into the barber chair upon arrival.
    ii. Expected number of customers waiting for haircut.
    iii. Effective arrival rate.
    iv. The time a customer can expect to spend in the barbershop.

c) A tax consulting firm has 3 counters in its offices to receive the people who have problems concerning their income and the sales tax. On an average 48 persons arrive in 8hrs a day. Each tax advisor spends 15 min an on average for a arrival of the arrival time follows a Poisson distribution and the service time follows exponential distribution.
    i. Find the average number of customer in the system.
    ii. Average waiting time of the customer in the system.
    iii. Average number of customers waiting the queue for service.
    iv. Average waiting time of the customers in the queue.
    v. How many hours each week a tax advisor spends performing his job.
    vi. Probability that a customer has to wait before he gets service.
    vii. Expected number of idle tax advisors at any specified time

Question Two

Using diagram, describe the following terms as used in Teletraffic engineering

a) Network of queues.
b) Open queuing networks
c) Closed queuing networks

Question Three

Using illustrative equations and diagrams:-

a) Discuss the Tandem queues theorem
b) Discuss the Jackson queues theorem

END